DESCRIPTION	N-m	Ft. Lbs.	In. Lbs.
LDP Pump Mounting Bolts	1	-	11
EVAP Canister/LDP Mounting Bracket-to-Body Nuts	-	-	250
EVAP Canister/LDP Mounting Bracket-to-Body Bolts	-	-	212

CCV SYSTEM

DESCRIPTION

The 4.0L 6-cylinder engine is equipped with a Closed Crankcase Ventilation (CCV) system and a fixed orifice valve (Fig. 1). The 2.4L 4-cylinder uses a PCV valve. Refer to PCV Valve for 2.4L information.

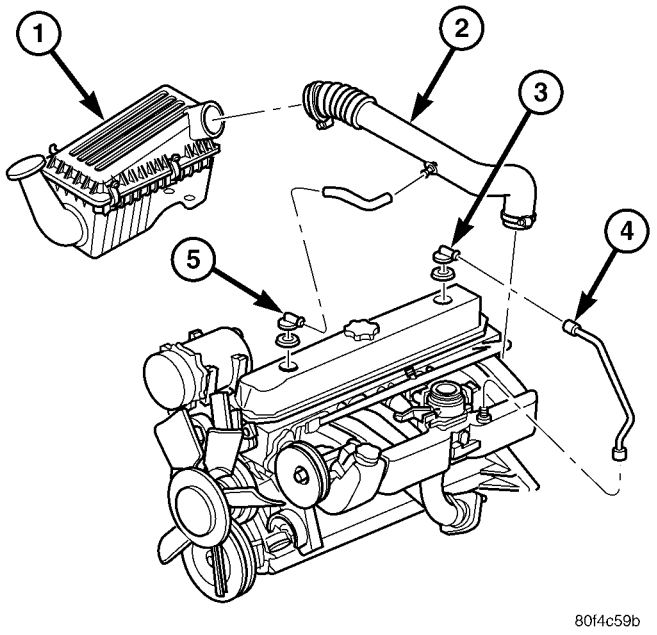

Fig. 1 CRANKCASE VENTILATION (CCV) FITTING - 4.0L

1 - AIR CLEANER ASSEMBLY
2 - AIR INTAKE TUBE
3 - FIXED ORIFICE FITTING
4 - VACUUM LINE
5 - AIR INLET FITTING

OPERATION

The CCV system performs the same function as a conventional PCV system, but does not use a vacuum controlled valve.

On 4.0L 6-cylinder engines, a molded vacuum tube connects manifold vacuum to the top of the cylinder head (valve) cover. The vacuum fitting contains a fixed orifice of a calibrated size. It meters the amount of crankcase vapors drawn out of the engine.

When the engine is operating, fresh air enters the engine and mixes with crankcase vapors. Manifold vacuum draws the vapor/air mixture through the fixed orifice and into the intake manifold. The vapors are then consumed during combustion.

DIAGNOSIS AND TESTING

TESTING / CLEANING

The Crankcase Ventilation (CCV) system performs the same function as a conventional PCV system, but does not use a vacuum controlled valve. A vacuum fitting containing a fixed orifice of a calibrated size is used. It meters the amount of crankcase vapors drawn out of the engine.

(1) Check each CCV system tube (line) for leaks, cracks, kinks or bends. Replace as necessary

(2) Disconnect each CCV tube.

(3) Remove fixed orifice fitting from valve cover.

(4) Blow compressed air through each tube, and the fixed orifice fitting. Check for blockage or restrictions. If cleaning is necessary, spray a soapy-type all-purpose cleaner into each component and blow out. After restriction is cleared, rinse out component with clear water. Blow water from component and install to vehicle. **To prevent damage to plastic components, never spray carburetor-type cleaner into any of the plastic tubes or the fixed orifice fitting. Never attempt to clean the fixed orifice fitting with a metal object as calibration could be affected. If fixed fitting cannot be cleared, replace it.**

PCV VALVE

DESCRIPTION

2.4L

The 2.4L 4-cylinder engine is equipped with a closed crankcase ventilation system and a Positive Crankcase Ventilation (PCV) valve.

This system consists of: